Project Initialization and Planning Phase

Date	01 October 2026
Team ID	SWTID-2026-6912
Project Title	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

FitBuddy is a web-based AI fitness planning application that generates personalized 7-day workout plans and goal-aligned nutrition or recovery tips. It uses Google Gemini models with a FastAPI backend, Jinja2-based HTML interface, and SQLite database support. Users can also submit feedback so that the generated plan can be refined according to their changing needs.

Project Overview	
Objective	To provide users with a structured and personalized 7-day fitness plan based on age, weight, fitness goal and preferred workout intensity, together with a complementary nutrition or recovery tip.
Scope	The project covers AI-based workout generation, nutrition/recovery tips, feedback-based plan updates, user and plan storage, a responsive web interface, API endpoints and an admin view of registered users and their plans.
Problem Statement	
Description	Users may need help creating structured routines that match their fitness goals and intensity preferences. Generic plans may not easily adapt when users provide new feedback or request changes.
Impact	A personalized, feedback-driven system can make fitness planning more structured and easier to refine while giving users quick access to goal-specific workout and nutrition guidance.
Proposed Solution	
Approach	Use Gemini 1.5 Pro for structured 7-day workout generation and feedback-based plan updating, and Gemini Flash for fast nutrition/recovery tips. FastAPI manages routing and processing, while SQLite with SQLAlchemy stores user and plan information.
Key Features	• Personalized 7-day workout plan • Nutrition/recovery tip generation • Feedback-based workout plan updates • User and plan storage • Admin view of original and updated plans • Responsive HTML/Jinja2 interface

Project Initialization and Planning Phase

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	System for application development, local server execution and API testing.	Development laptop/desktop; GPU not required for local application logic.
Memory	RAM for IDE, Python environment, local server and database.	8 GB or above
Storage	Disk space for source code, templates, SQLite database, dependencies and project documents.	At least 256 GB SSD recommended
Software		
Programming	Python development environment.	Python + pip
Backend	API and server framework.	FastAPI + Uvicorn
AI Integration	Generative AI services for workout and nutrition generation.	Google Gemini API / Gemini models
Frontend	User interface and dynamic rendering.	HTML, CSS, Jinja2
Database	Persistent storage of user details and original/updated plans.	SQLite + SQLAlchemy
Version Control	Source-code management and collaboration.	Git
Testing / API	Local browser testing and interactive API documentation.	Browser + FastAPI /docs

Project Workflow

Milestone	Activities	Expected Output
1. Model Selection and Architecture	Research and select the appropriate Gemini model; define application architecture; set up development environment.	Selected Gemini models, modular architecture and working development setup.
2. Core Functionalities Development	Develop workout generation, nutrition tips, feedback-based updates, user/plan storage and FastAPI backend processing.	Core AI functions, database operations and backend processing.
3. routes.py Development	Implement home, workout generation, feedback and admin routes and connect them to AI/database functions.	Working application routing and main application logic.
4. Frontend Development	Design responsive interface and create dynamic Jinja2 templates for input, results, feedback and admin view.	index.html, result.html, all_users.html and responsive UI.
5. Deployment	Prepare local deployment, configure environment variables, run the FastAPI server and verify the application.	Tested local application and deployment documentation.

Project Initialization and Planning Phase

Team Members and Work Allocation

S.No	Name	Assigned Work
1	**Pavithra M**	**Model Selection & Architecture** Research Gemini model requirements; support selection of Gemini 1.5 Pro and Gemini Flash; define the modular FastAPI architecture and development environment setup.
2	**Manisha S**	**Core AI Functionality** Develop workout plan generation using Gemini 1.5 Pro, nutrition/recovery tip generation using Gemini Flash, and feedback-based workout plan updating.
3	**Monica G**	**Backend, Routes & Database** Implement FastAPI input processing, Pydantic validation, routes.py logic and SQLite/SQLAlchemy operations for saving, retrieving and updating users and plans.
4	**Monisha Anandan**	**Frontend & Jinja2 Templates** Design the user interface and develop index.html, result.html and all_users.html with responsive CSS, dynamic Jinja2 rendering and feedback/admin views.
5	**Mridula M**	**Deployment, Testing & Documentation** Prepare local deployment, configure environment variables, verify the main user flows and API endpoints, record testing results and maintain project documentation.

Team Responsibility Summary

Member	Primary Responsibility	Related Project Area
Pavithra M	AI model research and architecture	Milestone 1
Manisha S	Gemini-powered core functionality	Milestone 2
Monica G	FastAPI routes, validation and database	Milestones 2 & 3
Monisha Anandan	Frontend UI and Jinja2 templates	Milestone 4
Mridula M	Deployment, testing and documentation	Milestone 5

Project Outcome

FitBuddy combines Gemini-based personalized workout generation, nutrition/recovery guidance, feedback-driven plan refinement, persistent storage and an interactive web interface. The team allocation above follows the project workflow and assigns each member a distinct primary area while allowing collaboration across integrated components.